# Algebraic Topology: Selected Problems

This document contains a curated list of foundational problems in point-set and algebraic topology, focusing on homotopy equivalences, retractions, locally connected spaces, fundamental groups, and deformation retracts.

**Question 1:** Let $A$ and $X$ be topological spaces, and let $i : A \to X$ be a homotopy equivalence. Does this imply that $A$ is a retract of $X$?

**Question 2:** Let $X$ be a topological space. Suppose that for every $x \in X$ and every neighborhood $U$ of $x$, there exists a connected neighborhood $V$ of $x$ such that $x \in V \subseteq U$. Show that local connectedness is preserved under retractions, and verify whether $X = I \times \{0\} \cup A \times I$ (where $A = \{0, 1, \frac{1}{2}, \dots\}$) is locally connected.

**Question 3:** Let $X = I \times \{0\} \cup A \times I$, where $A = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$. Does there exist a retraction $r : I \times I \to X$?

**Question 4:** Does a local homeomorphism necessarily imply a covering map?

**Question 5:** Find a space $X$ and base points such that $\pi_1(X, x_0) \neq \pi_1(X, \tilde{x}_0)$.

**Question 6:** Let $L = \bigcup_{n=0}^{\infty} L_n$, where $L_0 = I \times \{0\}$ and $L_n = \left\{\left(x, \frac{1}{n}\right) \mid x \in [0, 1]\right\}$ for $n \in \mathbb{N}$. Does there exist a strong deformation $F : L \times I \to L$ satisfying:

$$\begin{aligned} F(s, 0) &= s \\ F(s, 1) &= (1, 0) \\ F((1, 0), t) &= (1, 0) \quad \forall t \in I? \end{aligned}$$

**Question 7:** Does a strong deformation retract imply a deformation retract? What about the converse? Consider the space $L$ from the previous problem and investigate explicit homotopies for its deformation retraction onto the point $(1, 0)$.